



RESEARCH ARTICLE

A curated collection of transcriptome datasets to study the transcriptional response in blood and nasal samples following viral respiratory inoculation and vaccination

[version 1; peer review: 1 approved with reservations]

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Abstract

Background

Our understanding of the human immune system's response to viral respiratory tract infections (VRTIs) and vaccines, including the molecular mechanisms and correlates of protection, remains incomplete. Extensive transcriptomic data from inoculation and vaccination studies have been deposited in publicly available databases. However, these studies are often separate and difficult to locate.

Methods

To bridge this research gap, we have systematically searched and reviewed publicly available datasets from NCBI Gene Expression Omnibus (GEO), Archive of Functional Genomics Data (ArrayExpress), Immunology Database and Analysis Portal (ImmPort), and Google

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1

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1. Luiz Gustavo Gardinassi , University of Sao Paulo, Ribeirão Preto, Brazil

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using targeted queries. Inoculation study queries included terms related to humans, blood, PBMCs, nasal, respiratory challenges, and respiratory viral inoculations; while vaccination study queries focused on humans, blood, PBMCs, transcriptomes, and respiratory viral vaccines or vaccinations.

Results

This collection includes 18 datasets from inoculation of 5 respiratory viruses: H1N1, H3N2, RSV, HRV, and SARS-CoV-2 (14 from blood and 4 from nasal swabs) with 429 participants (ages 18 to 55 years) and 37 datasets from vaccination of influenza and COVID-19 with 2,084 participants (ages 0.5 to over 89 years). The duration and number of post-immunization time points range from 14 days before to 28 days after inoculation (1 to 20 time points) and from 28 days before to 360 days after vaccination (1 to 13 time points).

Conclusion

We provide a curated compendium of public gene expression data repositories for researchers to reanalyze transcriptomes from human whole blood, peripheral blood mononuclear cells (PBMCs), and nasal swab samples. This will facilitate studies of transcriptional responses to respiratory viral inoculation or vaccination.

Keywords

Transcriptomics, Bioinformatics, Vaccination, Inoculation, Respiratory viral infection, Influenza viruses, COVID, Respiratory syncytial viruses (RSV), Human rhinoviruses (HRV) Whole Blood, PBMC.

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Introduction

Viral respiratory tract infections (VRTIs) are widespread globally, causing significant illnesses and hospitalizations each year.¹ The importance of studying VRTIs has grown in light of the global COVID-19 pandemic.² VRTIs involve a range of viruses, including respiratory syncytial viruses (RSV), human rhinoviruses (HRV), and influenza and coronaviruses.³ Most VRTIs lack both effective antiviral therapies and approved vaccines.⁴

In-depth investigation of human immune responses to inoculation and vaccination is critically needed. To facilitate further research, we have compiled a curated dataset collection, which includes transcriptomic data from the blood and nasal swabs of volunteers who underwent inoculation and vaccination, specifically for influenza, COVID-19, RSV, and HRV.

Our curated dataset collection is divided into two categories: inoculation studies and vaccination studies. The inoculation collection includes 18 datasets with 429 participants, while the vaccination collection comprises 37 datasets with 2,084 participants. All datasets were sourced from GEO, ImmPort Shared Data, and the ArrayExpress Collection at EMBL-EBI.⁵⁻⁷ These datasets have been organized into a publicly accessible format for download and further analysis.

Methods

The datasets were gathered using specific search queries across GEO, ArrayExpress, ImmPort, and Google search. The queries for inoculation studies included terms related to humans, blood, PBMCs, nasal, respiratory challenges, and viral inoculations. The vaccination study queries followed similar criteria, focusing on humans, blood, PBMCs, transcriptomes, and vaccines or vaccinations. The search queries used for GEO, ArrayExpress, and Google search to find inoculation studies are as follows:

GEO and ArrayExpress Inoculation Search Query

("humans"[MeSH Terms] OR "Homo sapiens"[Organism] OR Homo sapiens [All Fields]) AND (("blood"[Subheading] OR "blood"[MeSH Terms] OR blood [All Fields]) OR PBMC [All Fields] OR PBMCs [All Fields]) AND respiratory [All Fields] AND (challenge [All Fields] OR "experimentally infected"[All Fields] OR inoculated [All Fields])

Google Inoculation Search Query

Homo sapiens AND (blood OR PBMC OR PBMCs) AND transcriptome AND respiratory AND (inoculation OR challenge OR "experimentally infected" OR inoculated)

The search queries used for GEO, ArrayExpress, and Google to find vaccination studies are as follows:

GEO and ArrayExpress Vaccine Search Query

("humans"[MeSH Terms] OR "Homo sapiens"[Organism] OR Homo sapiens [All Fields]) AND (("blood"[Subheading] OR "blood"[MeSH Terms] OR blood [All Fields]) OR PBMC [All Fields] OR PBMCs [All Fields]) AND respiratory [All Fields] AND (("vaccination"[MeSH Terms] OR inoculation [All Fields]) OR ("vaccines"[MeSH Terms] OR vaccine [All Fields]) OR ("vaccines"[MeSH Terms] OR vaccines [All Fields]) OR ("vaccination"[MeSH Terms] OR vaccination [All Fields]))

Google Vaccine Search Query

Homo sapiens AND (blood OR PBMC OR PBMCs) AND transcriptome AND respiratory AND (vaccine OR vaccines OR vaccination)

For **ImmPort**, vaccine studies were located using the following filters:

- Species: Homo sapiens
- Research Focus: Vaccine Response
- Condition or Disease: COVID-19 AND Influenza

Results

The majority of datasets obtained were derived from human blood, PBMCs, and nasal swabs, generated using Illumina or Affymetrix platforms or through RNA sequencing. Every dataset identified through this search was carefully curated manually. This involved thoroughly reviewing the dataset descriptions, examining the study designs, and reading

through the related original articles on PubMed. Ultimately, we only included studies that involved human whole blood, PBMCs, and nasal swabs linked to VRTI inoculation or vaccination for our dataset collection. According to these criteria, we retained 18 inoculation datasets and 37 vaccine-related datasets.

The data selection process for inoculation is summarized in [Figure 1a](#). Using GEO and ArrayExpress inoculation query, 108 studies were found with Homo sapiens as the primary organism. Google search returned 92 inoculation studies. The data selection process for vaccination is summarized in [Figure 1b](#). GEO and ArrayExpress Vaccine query identified 82 studies where Homo sapiens is the top organism. Google search produced 89 vaccination studies. ImmPort search yielded 99 vaccination studies.

[Figure 2](#) illustrates the distribution of participants over time in 18 inoculation studies, involving 429 participants aged 18 to 55 years, across five respiratory viruses: H1N1, H3N2, RSV, HRV, and SARS-CoV-2. There is a notable peak on day 0. The duration and number of post-immunization time points range from 14 days before to 28 days after inoculation, spanning 1 to 20 time points. Notably, 56 participants from four studies had both blood and nasal samples, and they were counted separately. Subsequent time points show smaller but consistent sample collections, with varying contributions from different pathogens. H1N1, H3N2, and HRV are the most frequently sampled pathogens, while SARS-CoV-2 and RSV contribute fewer samples to the dataset.

[Figure 3](#) summarizes the distribution of 2,084 participants, aged 0.5 to over 89 years, over time in 37 vaccination studies covering COVID-19 and influenza vaccines and their types. The peak occurred on day 0, with participant recruitment enriched during the first week. The duration and number of post-immunization time points range from 28 days before to 360 days after vaccination, spanning 1 to 13 time points.

The complete list of inoculation datasets included in our collection can be found in [Table 1](#).

We have collected transcriptomes from 18 cohorts across 10 inoculation studies, totaling 429 participants. This collection covers inoculations of five respiratory viruses: H1N1, H3N2, RSV, HRV, and SARS-CoV-2. The time span ranges from -14 to 28 days before and post for inoculation. The sample source includes blood (n=14) and nasal swabs (n=4). The transcriptomic data types include microarrays from Illumina (n=2) and Affy (n=11); bulk RNA-seq (n=5) ([Figure 4](#)).

The complete list of vaccination datasets included in our collection can be found in [Table 2](#).

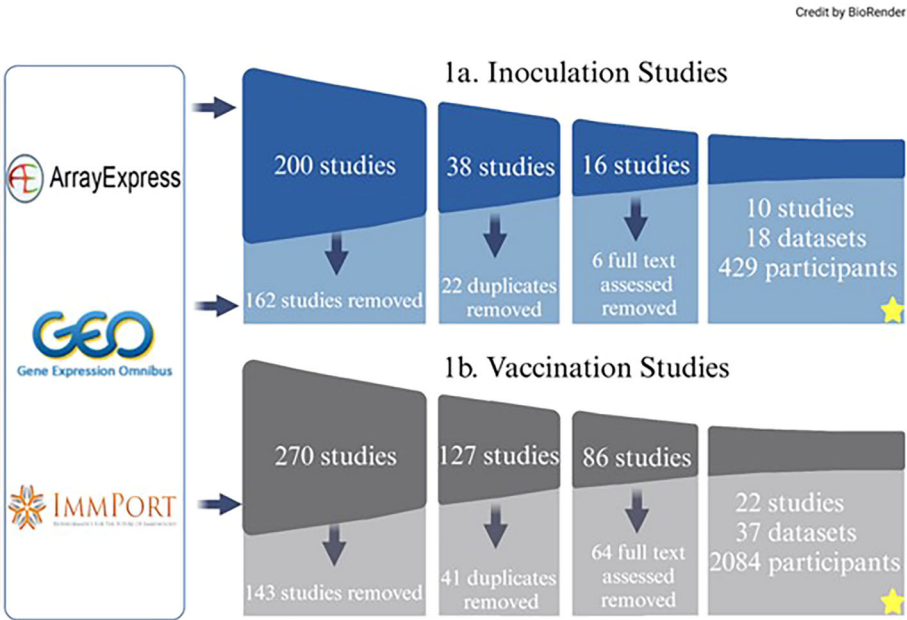


Figure 1. Flowchart illustrating construction of inoculation (a) and vaccination (b) dataset compendium.

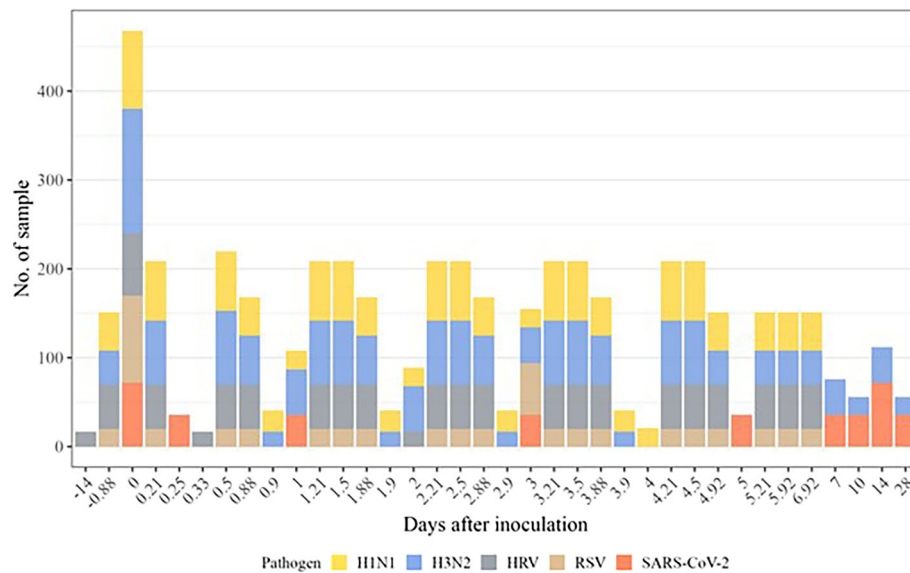


Figure 2. A collection of datasets covering transcriptional responses to inoculation over time across varying viruses. Histogram of the time points pre- and post-inoculation available in our compendium. Each virus is indicated by a different color. The height of the bars represents the number of participants with available gene expression data.

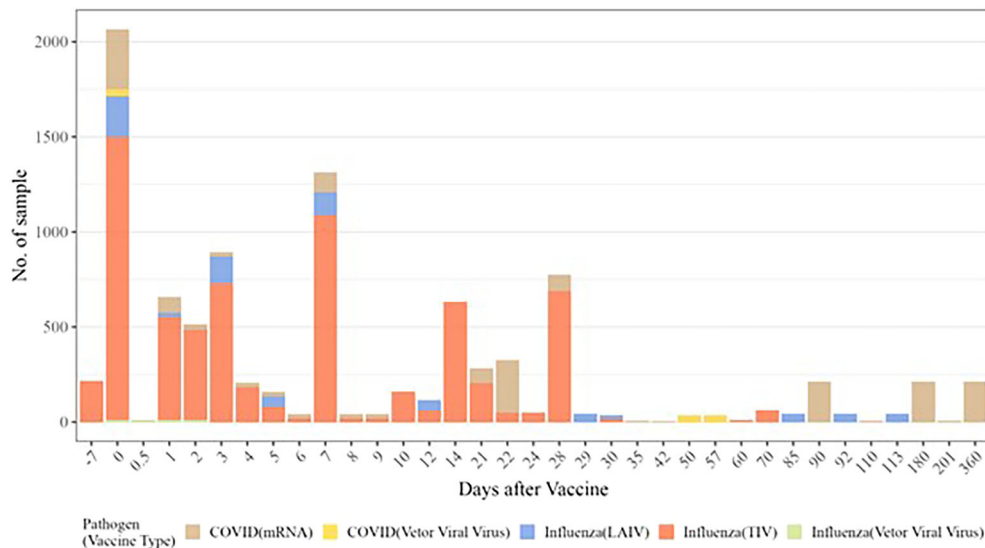


Figure 3. A collection of datasets covering transcriptional responses to vaccination over time across COVID-19 and influenza. Histogram of the time points pre- and post-vaccination is available in our compendium. Each vaccine and type are indicated by a different color. The height of the bars represents the number of participants with available gene expression data.

We have gathered transcriptomes from 37 cohorts across 22 vaccine studies, totaling 2084 participants. This collection focuses on studying Influenza and COVID-19 vaccines. The time span ranges from -28 to 360 days. The data source includes blood (n=34) and nasal swabs (n=3). The transcriptomic data types include microarrays from Illumina (n=13) and Affy (n=7); bulk RNA-seq (n=13) and scRNA-seq (n=4) (Figure 5). Vaccine types: TIV (n=22) and LAIV (n=6) were used for influenza vaccination, mRNA vaccine (n=7) was used for COVID-19 vaccination, and vector adenovirus (n=2) was used for both influenza and COVID-19, one of each.

Table 1. Complete list of inoculation datasets.

Pathogen	Sample type	Sample size	Profiling platform	Time point	Accession number	Citation
HRV	Peripheral blood	50	Microarray (Affymetrix)	-0.88, 0, 0.21, 0.5, 0.88, 1.21, 1.5, 1.88, 2.21, 2.5, 2.88, 3.21, 3.5, 3.88, 4.21, 4.5, 4.92, 5.21, 5.92, 6.92	GSE73072	8
RSV		20				
H3N2		38				
H1N1		43				
HRV	Peripheral blood	20	Microarray (Affymetrix)	0	GSE17156	9
RSV		20				
H3N2		17				
H3N2	Peripheral blood	17	Microarray (Affymetrix)	0, 0.21, 0.50, 0.88, 1.21, 1.50, 1.88, 2.21, 2.50, 2.88, 3.21, 3.50, 3.88, 4.21, 4.50	GSE30550	10
H3N2	Whole blood	11	Microarray (Illumina)	0,0.50,1,2	GSE61754	11
SARS-CoV-2	Blood and nose swabs	36	RNA-seq	0, 0.25, 14, 28(blood) 0,1,3,5,7,10,14 (nose swabs)	E-MTAB-12993	12
H3N2	Blood and nose swabs	20	RNA-seq	0, 1, 2, 3, 7, 10, 14, and 28(blood) 0,1, 2,3,7,14(nose swabs)	EGAD50000000956	
HRV	Nose swabs	17	Microarray (Affymetrix)	-14,0.33, 2.00	GSE11348	13
H1N1	Whole blood	21	Microarray (Illumina)	0, 1, 2, 3, 4	GSE90732	14
RSV	Nose swabs	58	RNA-seq	0,3	GSE155237	15
H1N1	PBMC	24	Microarray (Affymetrix)	0, 0.21, 0.50, 0.90, 1.21, 1.50, 1.90, 2.21, 2.50, 2.90, 3.21, 3.50, 3.90, 4.21, 4.50	GSE52428	16
H3N2		17				

PBMC: peripheral blood mononuclear cells.

Time point: Unit is day. 0 is the baseline, negative number means days before inoculation, and positive means days after inoculation.

Accession number: datasets from GEO start with "GSE", datasets from ArrayExpress start with "E-MTAB", dataset from EGA starts with "EGAD".

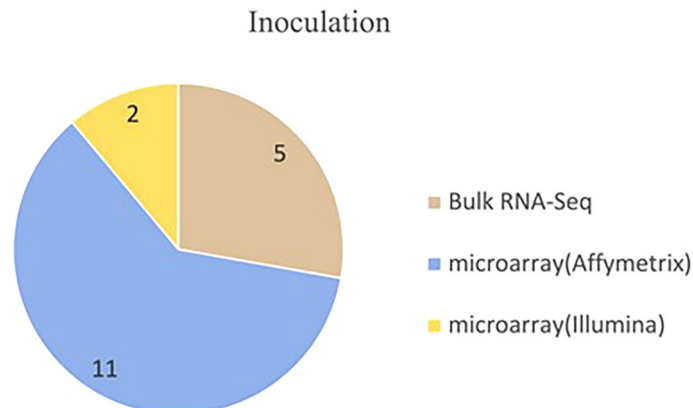
**Figure 4. Distribution of inoculation datasets across different profiling platforms.**

Table 2. Complete list of vaccine datasets.

Pathogen	Vaccine type	Sample type	Sample size	Profiling platform	Time point	Accession number	Citation
Influenza	TIV	PBMC	172	Microarray (Illumina)	0,2,7,28	GSE107990	17
Influenza	LAIV	Whole blood	20	Microarray (Illumina)	0,1,7,30	GSE52005	18
Influenza	TIV	Whole blood	17	Microarray (Illumina)	0,1,7,30		
Influenza	LAIV	PBMC	28	Microarray (Affymetrix)	0,3,7	GSE29619	19
Influenza	TIV	PBMC	28	Microarray (Affymetrix)	0,3,7		
Influenza	LAIV and inactivated	Blood	44	RNA-seq	0,3,7,29,85,92,113	GSE217770	20
Influenza	inactivated	PBMC	50	RNA-seq	0,1,3,7,21,22,24,28	GSE102012	21
Influenza	LAIV	Nasal epithelium	40	RNA-seq	0,3	GSE230494	22
Influenza	LAIV	Nasal cells	55	RNA-seq	0,5,12	GSE117580	23
Influenza	TIV	Nasal cells	62	RNA-seq	0,5,12		
Influenza	vetor viral virus	Whole blood	11	Microarray (Illumina)	0,0,5,1,2	GSE61754	11
Influenza	Inactivated	PBMC	14	RNA-seq	-7,0,1,2,3,4,5,6,7,8,9,10,21	GSE45764	24
Influenza	Inactivated	Whole blood	18	Microarray (Illumina)	-7,0,1,3,7,10,14,21,28	GSE30101/GSE48762	25
Influenza	Inactivated	PBMC	212	Microarray (Affymetrix)	0,3,7,14	GSE74817	26
Influenza	Inactivated	Whole blood	247	Microarray (Illumina)	0,1,3,14	GSE48024	
Influenza	Inactivated	Whole blood	91	Microarray (Illumina)	0	GSE41080	
Influenza	Inactivated	PBMC	5	RNA-seq	0,1,2,3,4,5,6,7,8,9,10	GSE45735	
Influenza	Inactivated	PBMC	60	Microarray (Illumina)	0,2,4,7,28	GSE59743/GSE95584	
Influenza	Inactivated	PBMC	27	Microarray (Affymetrix)	0,3,7	GSE29617/GSE29614	
Influenza	Inactivated	PBMC	64	Microarray (Illumina)	0,2,4,7,28	GSE59654	
Influenza	Inactivated	Whole blood	51	Microarray (Illumina)	0,2,7,28	GSE101709	
Influenza	Inactivated	PBMC	42	Microarray (Illumina)	0,4,7,28	GSE59635	
Influenza	Inactivated	Whole blood	44	Microarray (Illumina)	0,2,7,28	GSE101710	
Influenza	Inactivated	PBMC	63	Microarray (Affymetrix)	-7,0,1,7,70	GSE47353	
Influenza	LAIV	PBMC	28	Microarray (Affymetrix)	0,3,7	GSE29615	

Table 2. *Continued*

Pathogen	Vaccine type	Sample type	Sample size	Profiling platform	Time point	Accession number	Citation
influenza	Inactivated	Whole blood	123	Microarray (Affymetrix)	-7,0,1,3,7,10,14,21,28	SDY311/SDY312/SDY314/SDY315/SDY112	²⁷
Influenza	TIV	Whole blood	34	Microarray (Illumina)	0,7,14	SDY272/SDY648/SDY739/SDY819/SDY622	²⁸
influenza	TIV	Whole blood	65	RNA-seq	0,2,7,28	SDY1393	²⁹
Influenza	TIV	Whole blood	6	RNA-seq	1,3,60	SDY300	[*]
COVID	mRNA	PBMC	16	scRNA	0,7,21,28	GSE247917	³⁰
COVID	mRNA	Blood	23	RNA-seq	0,1,2,3,4,5,6,7,8,9	GSE190001	³¹
COVID	mRNA	PBMC	214	RNA-seq	0,22,90,180,360	GSE220682	³²
COVID	mRNA	Blood	6	scRNA	0,1,2,7,21,22,28,42	GSE171964	³³
COVID	mRNA	Blood	56	RNA-seq	0,1,7,21,22,28	GSE169159	
COVID	mRNA	PBMC	8	scRNA	28,35,60,110,201	GSE195673	³⁴
COVID	mRNA	PBMC	4	scRNA	0,7,30,90	GSE210229	³⁵
COVID	vector viral virus	Whole blood	36	RNA-seq	0,50,57	GSE228842	³⁶

PBMC: peripheral blood mononuclear cells; LAV: live attenuated influenza vaccine; TIV: trivalent inactivated influenza vaccine.

Time point: Unit is day. 0 is the baseline, negative number means days before vaccination, and positive means days after vaccination.

Accession number: datasets from GEO start with "GSE", datasets from ImmPort start with "SDY".

^{*}This data is available at ImmPort (<https://immport.org/shared/home>) under study accession SDY300: Healthy Human DC and monocyte subsets transcriptional regulations in response to Fluzone 2010-2011 and pneumococcal vaccinations.

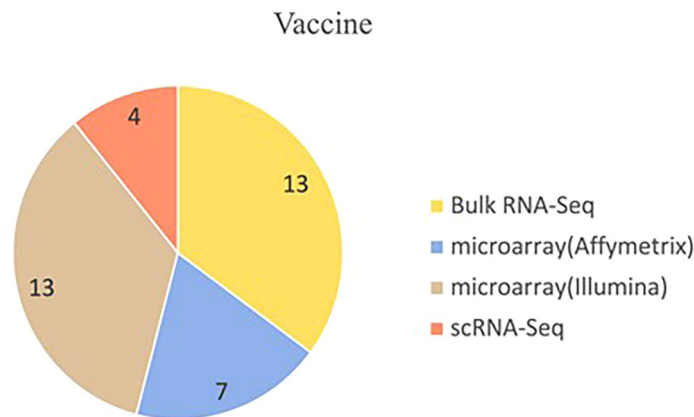


Figure 5. Distribution of vaccine datasets across different profiling platforms.

Ethics and consent

Ethical approval and consent were not required.

Data availability

All datasets in our curated collection are also accessible to the public on the NCBI GEO website at <https://www.ncbi.nlm.nih.gov/gds/> ArrayExpress at <https://www.ebi.ac.uk/biostudies/arrayexpress/studies> ImmPort at <https://www.immport.org/shared/home>, and the European Genome-phenome Archive (EGA)³⁷ at <https://ega-archive.org/datasets/> cited in the manuscript using its GEO number, ArrayExpress number, ImmPort study number or EGA study number. All these citation and accession numbers are listed in Table 1. The dataset can be found by searching for the accession number on the corresponding websites listed above. Note that downloading from ImmPort requires registering a personal account on their website.

Accession number

NCBI: Expression Omnibus (GEO): Transcriptome datasets of the transcriptional response in blood samples following HRV, RSV, H3N2 and H1N1 inoculation. Accession Number: GSE73072; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE73072>⁸

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets of the transcriptional response in blood samples following HRV, RSV and H3N2 inoculation. Accession Number: GSE17156; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE17156>⁹

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets of the transcriptional response in blood samples following H3N2 inoculation. Accession Number: GSE30550; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE30550>¹⁰

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets of the transcriptional response in blood samples following H3N2 inoculation. Accession Number: GSE61754; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE61754>¹¹

EMBL-EBI: ArrayExpress: Transcriptome datasets of the transcriptional response in blood and nasal samples following SARS-CoV-2 inoculation. Accession Number: E-MTAB-12993; <https://www.ebi.ac.uk/biostudies/arrayexpress/studies/E-MTAB-12993>¹²

European Genome-Phenome Archive (EGA): Transcriptome datasets to study the transcriptional response in blood samples following H3N2 inoculation. Accession Number: EGAD50000000956; <https://ega-archive.org/datasets/EGAD50000000956>¹² (Data available on request)

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in nasal samples following HRV inoculation. Accession Number: GSE11348; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE11348>¹⁴

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following H1N1 inoculation. Accession Number: GSE90732; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE90732>¹⁵

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in nasal samples following RSV inoculation. Accession Number: GSE155237; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE155237>¹⁵

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following H1N1 and H3N2 inoculation. Accession Number: GSE52428; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE52428>¹⁶

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following influenza vaccination. Accession Number: GSE107990; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE107990>¹⁷

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following influenza vaccination. Accession Number: GSE52005; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE52005>¹⁸

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following influenza vaccination. Accession Number: GSE29619; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE29619>¹⁹

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following influenza vaccination. Accession Number: GSE217770; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE217770>²⁰

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following influenza vaccination. Accession Number: GSE102012; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE102012>²¹

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in nasal samples following influenza vaccination. Accession Number: GSE230494; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE230494>²²

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in nasal samples following influenza vaccination. Accession Number: GSE117580; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE117580>²³

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following influenza vaccination. Accession Number: GSE61754; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE61754>¹¹

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following influenza vaccination. Accession Number: GSE45764; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE45764>²⁴

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following influenza vaccination. Accession Number: GSE30101 and GSE48762; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE30101> and <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE48762>²⁵

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following influenza vaccination. Accession Number: GSE74817; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE74817>²⁶

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following influenza vaccination. Accession Number: GSE48024; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE48024>²⁶

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following influenza vaccination. Accession Number: GSE41080; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE41080>²⁶

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following influenza vaccination. Accession Number: GSE45735; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE45735>²⁶

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following influenza vaccination. Accession Number: GSE59743 and GSE95584; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE59743> and <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE95584>²⁶

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following influenza vaccination. Accession Number: GSE29617 and GSE29614; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE29617> and <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE29614>²⁶

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following influenza vaccination. Accession Number: GSE59654; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE59654>²⁶

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following influenza vaccination. Accession Number: GSE101709; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE101709>²⁶

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following influenza vaccination. Accession Number: GSE59635; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE59635>²⁶

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following influenza vaccination. Accession Number: GSE101710; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE101710>²⁶

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following influenza vaccination. Accession Number: GSE47353; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE47353>²⁶

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following influenza vaccination. Accession Number: GSE29615; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE29615>²⁶

ImmPort: Transcriptome datasets to study the transcriptional response in blood samples following influenza vaccination. Accession Number: SDY311, SDY312, SDY314, SDY315 and SDY112; <https://www.immport.org/shared/study/SDY311>, <https://www.immport.org/shared/study/SDY312>, <https://www.immport.org/shared/study/SDY314>, <https://www.immport.org/shared/study/SDY315> and <https://www.immport.org/shared/study/SDY112>²⁷

ImmPort: Transcriptome datasets to study the transcriptional response in blood samples following influenza vaccination. Accession Number: SDY272, SDY648, SDY739, SDY819 and SDY622; <https://www.immport.org/shared/study/SDY272>, <https://www.immport.org/shared/study/SDY648>, <https://www.immport.org/shared/study/SDY739>, <https://www.immport.org/shared/study/SDY819> and <https://www.immport.org/shared/study/SDY622>²⁸

ImmPort: Transcriptome datasets to study the transcriptional response in blood samples following influenza vaccination. Accession Number: SDY1393; <https://www.immport.org/shared/study/SDY1393>²⁹

ImmPort: Transcriptome datasets to study the transcriptional response in blood samples following influenza vaccination. Accession Number: SDY300; <https://www.immport.org/shared/study/SDY300>

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following COVID vaccination. Accession Number: GSE247917; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE247917>³⁰

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following COVID vaccination. Accession Number: GSE190001; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE190001>³¹

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following COVID vaccination. Accession Number: GSE220682; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE220682>³²

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following COVID vaccination. Accession Number: GSE171964 and GSE169159; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE171964> and <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE169159>³³

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following COVID vaccination. Accession Number: GSE195673; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE195673>³⁴

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following COVID vaccination. Accession Number: GSE210229; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE210229>³⁵

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in blood samples following COVID vaccination. Accession Number: GSE228842; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE228842>³⁶

NCBI: Gene Expression Omnibus (GEO): Transcriptome datasets to study the transcriptional response in nasal samples following HRV inoculation. Accession Number: GSE11348; <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE11348>¹³

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The data compendium described in this manuscript can be a useful community resource. There are a few issues that need to be better explored.

- Define inoculation studies as controlled human challenges and the advantages of such experimental approach to understand host transcriptional responses in the introduction section. What are the main differences compared to natural infections? How are they more related to vaccine studies?

- Although the compendium gathered the studies into the same framework, it is unclear how many studies are ready to be used in statistical and other analysis and those that need additional processing before moving forward to statistical and functional analysis, such as filtering and normalization. This can have a major influence depending on the computational resources from users.

Is the work clearly and accurately presented and does it cite the current literature?

Yes

Is the study design appropriate and is the work technically sound?

Yes

Are sufficient details of methods and analysis provided to allow replication by others?

Yes

If applicable, is the statistical analysis and its interpretation appropriate?

Not applicable

Are all the source data underlying the results available to ensure full reproducibility?

Yes

Are the conclusions drawn adequately supported by the results?

Partly

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: Systems immunology, transcriptomics, metabolomics applied to the study of infectious and parasitic diseases.

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

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